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I Semester M.C.A. (Two Years Course) Examination, Aug./Sept. 2021
(CBCS) (2020-2021 and Onwards)
COMPUTER SCIENCE
1MCA 1 : The Art of Programming

Time : 3 Hours Max. Marks : 70

PART – A

Note : Answer **any five**. **Each** question carries **six** marks : **(5×6=30)**

1. What is complexity of an algorithm ? What is the best case, worst case and average case complexity ?
2. Write an algorithm to generate fibonacci series upto the number n.
3. Write an algorithm to reverse the digits of an integer. Check your algorithm for the input 1234.
4. What is the advantage of Binary search ? Write the binary search algorithm.
5. Write a note on control structures in C high level language.
6. How are matrices stored using row major representation ? Represent the following matrix using row-major order.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}_{4\times 3}.$$

How does the compiler find the location of a_{ij} th element ?

7. What is pass by value and pass by reference ? Explain with suitable example.
8. What is a structure ? Write a structure for the members : name, gender and salary. How are the members of the structure accessed in C high level language ?

P.T.O.

PART – B

Note : Answer **any four**. **Each** question carries **10** marks : **(4×10=40)**

9. What is 2-way merge sort algorithm ? Given the following two sets, merge the elements using 2-way merge sort

   List 1 : 3 4 6 9 11 14 23

   List 2 : 1 2 5 7 8 13 17 19

10. Write selection sort algorithm. Sort the following number using selection sort :

    7 1 4 3 9 2 0 8

    show the numbers after each pass.

11. Write an algorithm to multiply two matrices.

12. Write an algorithm to remove duplicates from a given list. Trace your algorithm for the following list :

    1 2 2 4 5 6 7 7 8 9 9 9

13. Write an algorithm to find a sub text in a given text. What is pattern searching ? How do you search for a key word in a given text ?

14. Write a C program to find the maximum number in a given list. Trace your algorithm for the following list :

    {4, 6, 3, 2, 9, 8, 13, 5}

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I Semester M.C.A. (2 Years Course) Examination, July 2022
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
1 MCA 1 : Art of Programming

Time : 3 Hours | Max. Marks : 70

Instructions : 1) *Part – **A** : Answer **any five** questions. **Each** question carries **6** marks.*
2) *Part – **B** : Answer **any four** questions. **Each** question carries **10** marks.*

PART – A

Answer **any five** questions : **(5×6=30)**

1. Write an algorithm to find : $1^2 + 3^2 + 5^2 + ... + n^2$.
2. Define Big oh notation. What is rate of growth, worst, best an average case complexity of an algorithm ?
3. Write an algorithm to convert the base of a number from base 10 to base 16.
4. Explain array of pointers.
5. Design an algorithm to find factorial of a given number using non recursive algorithm. Find factorial (0) and factorial (5) using your algorithm.
6. Explain algorithmic development for Histogramming.
7. Write an algorithm for binary search. Search 7 using your algorithm in a list {4, 6, 7, 8, 12, 14, 19}.
8. What is 2-way merging ? Merge {4, 6, 9, 10, 12, 15} and {1, 2, 3, 5, 8, 11, 13, 14} using your algorithm.

PART – B

Answer **any four** questions from Part – B : **(4×10=40)**

9. What is hashing ? What is collision control mechanism ? Hash the following numbers :
   {1, 2, 6, 9, 0, 4, 3, 10, 12}.
10. Explain formatted input and output in C.

P.T.O.

11. Discuss design and development of algorithm to compute Fibonacci numbers.

12. Write an algorithm to sort the numbers using selection sort. Sort the numbers {1, 0, 6, 5, 4, 9, 2, 3} using your algorithm.

13. Write an algorithm to remove duplicates in an array. Test your algorithm for the list {4, 5, 5, 6, 7, 7, 9, 12, 13, 13}.

14. Explain algorithm for multiplication of matrices. Multiply $A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \end{pmatrix}_{2\times3}$ and $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}_{3\times2}$.

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JP – 763

I Semester M.C.A. (Two Years Course) Examination, July 2023
(CBCS Scheme)
(2020 – 2021 and Onwards)
1MCA1 : THE ART OF PROGRAMMING

Time : 3 Hours Max. Marks : 70

PART – A (5×6=30)

Answer **any five** of the following :

1. What is an algorithm ? What is complexity of an algorithm ? Define Big Oh notation.
2. Write an algorithm to find fibonacci series upto n.
3. Write an efficient algorithm to find GCD of two numbers. Find GCD of (12, 5) and (54, 42).
4. Given a 'for' control structure and 'while' control structure which one do you use ? Explain your answer for reversing a number.
5. Write an algorithm to sort numbers using selection sort. Trace your algorithm to sort the following list of elements.
   {6, 3, 9, 8, 4, 5, 0, 1, 2, 7}.
6. Write an algorithm to find square root of a number.
7. What is the difference between a structure and a union ? Declare a structure for an organization with 100 employees. The employees name, age, identity-number may be stored.
8. What is hashing ? When does collission happen ? Given the following elements, store the elements in a hash table of size 10 : (use any of the hash function and collission handling algorithm).
   {6, 3, 13, 24, 36, 12, 4, 9}.

PART – B (4×10=40)

Answer **any four** of the following :

9. What is call by value, and call by reference ? Give an example. What is a recursion ?
10. By partitioning the set, sort the following elements.
    {4, 0, 9, 6, 8, 2, 7, 1, 5, 3}.
    Write merge-sort algorithm.

P.T.O.

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11. a) Write an algorithm to generate Pseudo random number. 5

    b) Write an algorithm to convert base 10 number to base 8 number. 5

12. Write an efficient algorithm to find a^n where n is a large number.

13. Write an algorithm to remove duplicates in an array. Trace your algorithm for the following :
    {13, 14, 14, 15, 15, 15, 16, 17, 18, 18, 19}.

14. Write a short note on the following :

    a) Two-way merging. 5

    b) Pattern matching algorithm. 5

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I Semester M.C.A. (Two Years Course) Examination, June/July 2024
(CBCS Scheme) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA1 : The Art of Computer Programming

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any five** questions from – Section **A**.*
*2) Answer **any four** questions from – Section **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Describe the different types of analysis of algorithm.
2. Write an algorithm to find ${}^{n}C_{r}$ using a function to find factorial of a number.
3. Explain break and continue statements by taking an example for each.
4. Explain the different looping construct used in C.
5. Write a C program to add two matrices.
6. Explain the procedure to reverse the elements of an array so that they appear in the reverse order by taking the elements {4, 3, 1, 0, 6, 9, 2, 5}.
7. How the elements of an array A = { 90, 50, 10, 40, 30} are sorted using insertion sort ? Write algorithm.
8. Write an algorithm to sort the numbers using bubble sort. Use the following elements to test your algorithm {9, 5, 6, 3, 1, 7, 0}.

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SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. a) Write an algorithm to find a^x where x is a large number. **5**
   b) Write an algorithm to reverse given integer. Test your algorithm to reverse 456. **5**
10. a) Define big-oh notation. Prove that f(n) = O (g(n)) by taking an example. **5**
    b) C program to input a 4-digit number and find the sum of its digits using do while loop. **5**

P.T.O.

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I Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
Paper – 1MCA1 : The Art of Programming

Time : 3 Hours Max. Marks : 70

Instruction : *Answer* ***any five*** *from Part –* ***A*** *and* ***any four*** *from Part –* ***B****.*

PART – A

Answer **any five** of the following. **Each** carries **six** marks. **(5×6=30)**

1. What is an algorithm ? What are the qualities of a good algorithm ?
2. What are the standard notations used to describe the asymptotic behavior of running time functions ? Give an account of standard functions.
3. Write both recursive and non-recursive algorithms to find the sum of the digits of an integer.
4. Write a C program to find whether a given date is valid or not [day, month and year are the inputs].
5. Write a C program to convert a decimal integer to any number system among Binary, Octal number systems.
6. Write an algorithm to find whether integer values in an integer array are unique or not.
7. Present selection sort algorithm and write C code for it.
8. Write a C program to search for a substring in a given string.

PART – B

Answer **any four** of the following. **Each** carries **ten** marks. **(4×10=40)**

9. a) Write algorithms to generate n members of the Fibonacci series. **5**
   i) Without using array of integers
   ii) Using array of integers.

   b) Write a C program to find the sum of the digits of a given integer till the sum reduces to a single digit. **5**

P.T.O.

10. a) Write an algorithm for sorting each row of a matrix and provide a C program for it. 5

    b) Write a C program to segregate prime numbers and composite numbers from a matrix of integers into two one-dimensional arrays. 5

11. a) Write an algorithm to find the number of times of occurrences of each number in a list of numbers, and display each number and the number of times it occurs. 5

    b) Describe the Naive method of computing the GCD of two given integers, and write a C program to implement the method. 5

12. a) Develop an algorithm to compute the prime factors of a given integer. 5

    b) Write a C program to remove the duplicate values in an array of integer values. 5

13. a) Write a C program to search for a string in a list of strings and also find its frequency of occurrence. 5

    b) Describe the procedure of sorting by insertion and present the complete algorithm for sorting by insertion. 5

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14. Write functions to illustrate the following : 10

    a) Finding the largest of four integers

    b) Break and continue statements in C

    c) Swapping two one-dimensional arrays

    d) Comparison of two strings.
